Dusty beginnings

About 4.6 billion years ago, a cloud of dust and gas began to fall in on itself. As it collapsed, it started to spin, forming a dense disk shape.

The solar system

The solar system is our home. A neighborhood of planets, moons, asteroids, and comets that dance in a cosmic ring around our star, the sun. And while the sun keeps everything going around it, locked in its gravity, it also holds the key to our solar system's future.

What next?

At the moment, our solar system is pretty stable. However, in about five billion years' time, the sun will run out of hydrogen and begin to burn helium. When this happens, it will swell up and cool down until it turns into a red giant—a type of star entering a late stage of its life—gobbling up Mercury, Venus, and maybe even the Earth!

Goldilocks zone This is the area of our solar system where water and life can exist. Like the porridge that Goldilocks eats, the conditions are perfect—it's neither too far from nor too close to the sun.

When the sun expands, this zone will move away from Earth. Eventually, some of the gas giants and their moons will be in the Goldilocks zone.